

STUDENT COURSE GUIDE

CESE323 ERC1.0

Cloud Security Envisioning & Strategy

Microsoft Defender for Cloud, Microsoft Entra, CSPM, workload protection, AI security, and operational readiness

Course code	CESE323
Release	ERC1.0
Guide type	Student Course Guide
Delivery model	One-day instructor-led workshop with optional self-paced extension
Primary platforms	Microsoft Azure, Microsoft Defender for Cloud, Microsoft Defender portal, Microsoft Entra ID, Microsoft Learn
Audience	Cloud security engineers, administrators, architects, SecOps analysts, technical leads, and decision-makers
Prepared for	Skunkworks Academy courseware development

Student promise

By the end of this course, you should be able to connect strategic cloud security priorities to practical Microsoft Defender for Cloud posture management, workload protection, AI security, identity governance, and remediation decisions.

COURSEWARE

Document Control

Use this page to confirm you are using the correct student guide for the course delivery.

Field	Value
Document title	CESE323 Student Course Guide
Course title	Cloud Security Envisioning & Strategy
Release	ERC1.0
Intended audience	Students and delegates attending the workshop or self-paced companion course
Primary source basis	Workshop syllabus, Microsoft Learn collection, and CESE323 pre-course lab readiness requirements
Document purpose	Provide students with the course context, expectations, agenda, lab readiness requirements, learning path, assessment approach, and post-course actions

Important

This student guide is not a replacement for the lab guide. The lab guide contains step-by-step exercise instructions, evidence capture requirements, and any environment-specific details provided by the instructor.

GUIDE MAP

How to Use This Guide

Read this guide before class, keep it open during the workshop, and return to the checklists after the labs.

Section	What it helps you do
1. Course overview	Understand the course purpose, scope, and expected outcomes.
2. Learning outcomes	Know what you should be able to explain, configure, review, and recommend.
3. Audience and prerequisites	Confirm that your baseline skills and accounts are ready.
4. Course journey and agenda	Follow the one-day learning flow from strategy to operational review.
5. Lab environment and safety	Understand lab permissions, data handling, change control, and cleanup rules.
6. Workload protection tracks	Prepare for the selected Defender workload modules.
7. Assessment and evidence	Know how participation, lab evidence, and scenario work are evaluated.
8. Study workflow	Plan pre-course, in-class, and post-course learning activities.
9. Glossary	Review key terms used throughout the course.

Appendices

Use checklists for readiness, portals, lab naming, cleanup, and notes.

SECTION 1

Course Overview

CESE323 is designed to bridge strategic cloud security planning and hands-on Microsoft Defender for Cloud implementation.

CESE323 ERC1.0 - Cloud Security Envisioning & Strategy helps students translate cloud security priorities into practical posture management, workload protection, threat reduction, and operational readiness activities. The workshop is structured around Microsoft Defender for Cloud, Microsoft Defender portal, Microsoft Entra ID, and related Microsoft Learn modules.

The course is intentionally outcome-led. Students do not only review product features; they learn how to use a cloud security architecture conversation to prioritize risk, identify misconfiguration, choose workload protections, plan remediation, and operate safely in lab or customer environments.

Course positioning

This course is best treated as a strategic and technical workshop. It supports security envisioning, architecture alignment, Defender for Cloud configuration, and operational review. It is not positioned as a full certification bootcamp, although several Microsoft Learn items align with SC-900, SC-100, SC-200, and AI security learning areas.

Dimension	Course scope
Primary focus	Cloud security posture, Microsoft Defender for Cloud, CSPM, workload protection, AI security, identity-aware controls, and remediation planning
Delivery duration	One day in instructor-led format, with optional self-paced Microsoft Learn extension
Hands-on depth	Portal-led configuration, policy assignment, posture review, selected Defender workload modules, and live/demonstration scenarios
Recommended lab model	Instructor-controlled Azure tenant or approved lab subscription with student-specific resource groups
Assessment model	Knowledge checks, lab evidence, scenario discussion, and optional capstone-style operational review

SECTION 2

Learning Outcomes

The student outcomes are grouped into strategic, technical, and operational capabilities.

- **Strategic alignment:** Define cloud security goals, prioritize risk reduction, and map technical actions to business outcomes.
- **Unified visibility:** Explain how Microsoft Defender for Cloud and Microsoft Defender portal contribute to a consolidated security view.
- **Threat reduction:** Identify misconfigurations, over-privileged identities, attack paths, vulnerable resources, and remediation opportunities.

- **Workload hardening:** Describe and select workload protections for storage, containers, applications, AI services, and related cloud workloads.
- **AI security awareness:** Explain common AI workload risks, including prompt injection, data leakage, unsafe model behavior, and identity/control-plane exposure.
- **Operational readiness:** Prepare for real-world rollout by considering permissions, costs, policies, change windows, rollback, and post-course cleanup.

After this course, you should be able to...	Evidence of achievement
Interpret Secure Score, recommendations, inventory, compliance, and workload protection coverage	You can explain which findings should be prioritized and why.
Select two suitable workload protection tracks for a given scenario	You can justify Defender for Storage, Containers, App Service, or AI Services based on risk and workload profile.
Apply least-privilege thinking to Defender and Azure access	You can identify when Reader, Contributor, Security Admin, or Owner is appropriate.
Describe AI-specific controls and data handling rules	You can explain why production data and confidential prompts must not be used in labs.
Build an action-oriented remediation summary	You can convert security findings into a prioritized operational plan.

SECTION 3

Audience, Prerequisites, and Student Readiness

Students should arrive with working access, tested MFA, and baseline knowledge of Azure, identity, and cloud security.

Audience group	Why this course is relevant
Cloud administrators	Understand how Defender for Cloud posture management and workload plans affect day-to-day Azure operations.
Security engineers	Use Defender findings, attack paths, and workload signals to prioritize remediation.
SecOps analysts	Connect Defender for Cloud posture and alerts to Microsoft Defender portal investigation workflows.
Solution architects	Map business risk, regulatory needs, and Zero Trust principles to practical Microsoft security controls.
Technical decision-makers	Understand cost, governance, rollout risk, and operational readiness before enabling protection plans.

Minimum knowledge baseline

Area	Expected baseline before class
Cloud fundamentals	Subscriptions, resource groups, regions, resources, identity, billing, and Azure RBAC.
Security fundamentals	Shared responsibility, Zero Trust, defense-in-depth, identity as the control plane, and common misconfigurations.
Microsoft Entra ID	Users, groups, roles, MFA, Conditional Access basics, and access management.
Microsoft Defender for Cloud	Basic navigation, Secure Score, recommendations, regulatory compliance, workload protection plans, and security posture management.
Azure workloads	Storage accounts, App Service, virtual machines, containers/AKS, and AI services where relevant.
AI security	Azure AI services, Microsoft Foundry or Azure OpenAI concepts, model deployments, prompts, and data leakage risks.

Required accounts and access

Requirement	Student action
Microsoft Learn account	Sign in and confirm you can access the assigned collection.
Azure portal access	Confirm you can sign in before class and can see the correct directory/tenant.
Microsoft Defender portal access	Required where incident, alert, or Defender XDR views are included.

Named account	Use your own assigned identity. Do not share admin credentials.
MFA	Configure and test before class.
Lab subscription/resource group	Confirm visibility and required role assignment before day one.
Modern browser	Use Microsoft Edge or Chrome. Avoid mobile-only access.

Readiness warning

Permissions are the most common blocker. Students should not discover during class that they cannot view the subscription, access Defender for Cloud, enable a plan, create resources, or capture lab evidence.

SECTION 4

Course Journey and One-Day Agenda

The course moves from strategy and governance to configuration, remediation, workload protection, and operational review.

Course part	Duration	Student focus
Part 1 - Strategic Initiation and Governance	90 minutes	Identify security goals, risk vectors, business priorities, architecture assumptions, and compliance gaps.
Morning break	10 minutes	Pause and prepare for hands-on Defender platform work.
Part 2 - Configuration Control and Workload Protection	120 minutes	Explore Microsoft Defender portal, CSPM, Security Exposure Management, attack paths, and workload protection selection.
Lunch break	45 minutes	Reset before practical configuration and remediation activities.
Part 3 - Practical Implementation and Remediation	75 minutes	Configure selected Defender components, review AI-specific security, and address rollout blockers.
Afternoon break	10 minutes	Pause before scenarios and environment review.
Part 4 - Hands-on Scenarios and Live Environment Review	75 minutes	Review scenarios, map threats to recommended mitigations, and conduct a guided operational review.

Learning flow

1. Start with business drivers: what must be protected, why it matters, and what risk reduction looks like.
2. Translate business drivers into cloud security posture priorities using Defender for Cloud and Microsoft security architecture concepts.
3. Review misconfiguration, over-privileged identities, attack paths, and exposure management signals.
4. Select two workload modules that match the learner environment and course priorities.
5. Configure, observe, or simulate Defender controls depending on permissions and approved lab scope.
6. Close with remediation planning, operational ownership, evidence capture, and cleanup steps.

Timing note

The Microsoft Learn modules associated with this course are longer than a single workshop day. The instructor-led day compresses the experience into priority discussions, demos, guided labs, and scenario reviews. Self-paced completion continues after class.

SECTION 5

Microsoft Learn Collection and Study Resources

The course references a curated Microsoft Learn collection and related Microsoft security learning paths.

Collection URL: [Cloud Security Envisioning & Strategy Workshop Syllabus collection](#)

Resource	Level	Role focus	Use in this course
SC-900T00-A: Introduction to Microsoft Security, Compliance, and Identity	Beginner	Student, business user	Pre-work or foundation refresh for security, compliance, identity, Azure, Microsoft 365, Entra, and Purview.
Introduction to security, compliance, and identity concepts	Beginner	Student, business user	Security concepts, shared responsibility, Zero Trust, encryption, GRC, and identity foundations.
Introduction to Microsoft Entra	Beginner	Student, business user	Identity types, authentication, MFA, password protection, and identity governance context.
Introduction to Microsoft security solutions	Beginner	Student, business user	Security Copilot, Azure infrastructure security, Defender for Cloud, and Defender XDR context.
Introduction to Microsoft Purview and privacy principles	Beginner	Student, business user	Data classification, sensitivity labels, DLP, insider risk, and privacy principles.
Manage security posture by using Microsoft Defender for Cloud	Intermediate	Administrator, security engineer	Hybrid/multicloud connection, CSPM, EASM, compliance, workload plans, and vulnerability management.
Mitigate threats using Microsoft Defender for Cloud	Intermediate	Security operations analyst	Threat mitigation, workload protection, Azure and non-Azure connection, and security posture management.
Secure AI solutions using Defender for Cloud and Microsoft Entra	Intermediate	Security engineer	AI workload security, Foundry protection, and identity controls for AI workloads.
SC-100T00-A: Microsoft Cybersecurity Architect	Advanced	Solution architect	Reference architecture, Zero Trust, CAF, WAF, MCRA, SecOps, identity, applications, data, and infrastructure.

SECTION 6

Lab Environment, Permissions, and Safety Rules

The course is hands-on only when the required permissions, budget approval, and safe lab scope are in place.

Lab model. The preferred student lab model is an instructor-controlled Azure tenant or approved training subscription with student-specific resource groups. Production or customer subscriptions should be treated as review-only unless explicit written approval, change control, rollback, and budget controls are in place.

Recommended role model

Role	Recommended student use	Risk control
Reader	Observation labs, posture review, inventory, recommendations, and Secure Score review.	Default for most students.
Contributor	Create lab resources and apply limited remediation in assigned resource groups.	Scope only to assigned resource group where possible.
Security Admin	Enable or manage selected Defender plans and policies where the lab requires it.	Use selectively and monitor changes.
Owner	Reserved for instructor, lab lead, or tightly controlled plan enablement scenarios.	Do not assign broadly.
Global Administrator	Not required for normal student lab work.	Reserve for emergency or instructor-controlled tenant management only.

Data handling rules

- Do not upload real customer, personal, HR, legal, financial, regulated, or confidential data into lab resources.
- Do not paste confidential business information, secrets, keys, credentials, source code, or proprietary customer data into AI prompts.
- Use only test files, sanitized data, instructor-provided artifacts, or deliberately disposable sample workloads.
- Capture screenshots only of allowed lab views. Do not expose secrets, tokens, keys, customer tenant identifiers, or sensitive resource names.
- Clean up lab resources and temporary role assignments when instructed.

Paid plan warning

Some Microsoft Defender for Cloud workload protection plans may create cost. Students must not enable paid plans outside the assigned subscription, resource group, or approved scope.

Approved live environment controls

Control	Required student understanding
Written approval	No live/customer subscription changes without explicit approval.
Named scope	Confirm which subscription, management group, or

	resource group is in scope.
Change window	Use a planned window for enabling plans, deploying agents, or changing policy.
Rollback plan	Know how changes will be disabled or reverted.
Break-glass access	Instructor or customer admin maintains emergency access.
Audit trail	Use named accounts only; avoid shared credentials.

SECTION 7

Workload Protection Tracks

The workshop selects two of four workload modules to keep the one-day experience focused and practical.

The syllabus allows the instructor or customer team to select **two out of four** workload protection modules. Selection should be driven by the audience, environment readiness, risk profile, and permission model.

Track	What it protects	Typical hands-on focus	Readiness requirement
Defender for Storage	Data layers, blob activity, malware upload risk, and suspicious access patterns.	Review storage posture, enable plan where approved, test sample upload/access scenarios.	Azure subscription, Defender for Cloud enabled, storage account, blob container, and test files only.
Defender for Containers	AKS and containerized workloads.	Review Kubernetes posture, registry/image risk, container recommendations, and sensor coverage where available.	AKS cluster, supported Kubernetes version, network access, permissions, and cost control.
Defender for App Service	Cloud-native applications and runtime environments.	Review App Service posture, recommendations, runtime protection concepts, and logging/diagnostics options.	App Service plan, sample app, isolated resource group, and selected Defender plan.
Defender for AI Services	Azure AI services, Microsoft Foundry, AI workload posture, and threat monitoring.	Enable or review AI workload plan, monitor Data and AI security dashboard, discuss prompt injection and data leakage risks.	Azure AI/Foundry resources, model endpoint, Owner or Contributor where enablement is required, and strict prompt/data safety rules.

How module selection is decided

- Start with business risk: data exposure, app security, container workloads, or AI adoption.
- Confirm lab readiness: permissions, subscription access, resource availability, and cost approval.
- Prioritize learning value: choose modules that reflect real operational decisions students will make after class.
- Avoid unsafe live changes: use demonstration mode where production risk or permissions block hands-on work.

Recommended default for mixed audiences

For a broad one-day student group, start with CSPM plus Defender for Storage or App Service. Add Defender for AI Services where Azure AI or Microsoft Foundry resources are available and data handling controls are clear.

SECTION 8

Practical Workflow During Labs

Use this repeatable workflow for posture review, workload protection, and remediation discussions.

Step	Student action	Expected output
1. Confirm scope	Check tenant, subscription, resource group, and role assignment.	Known safe working scope.
2. Observe posture	Review Secure Score, recommendations, inventory, regulatory compliance, workbooks, and exposure paths.	Initial risk view.
3. Prioritize risk	Identify high-impact findings, internet exposure, over-privileged identities, vulnerable resources, or missing plans.	Short risk-ranked list.
4. Select control	Choose an appropriate Defender plan, policy, or remediation path.	Justified technical control.
5. Configure or simulate	Perform the approved action in the lab or observe the instructor demonstration.	Configuration evidence or scenario notes.
6. Validate	Confirm recommendations, alerts, workbooks, or coverage views update as expected.	Validation evidence.
7. Document remediation	Capture the finding, impact, recommended action, owner, and next step.	Remediation summary.
8. Clean up	Remove resources, disable unneeded plans, and revoke temporary access as instructed.	Reduced cost and risk after lab.

SECTION 9

Assessment, Participation, and Evidence Expectations

The assessment model is designed to test applied reasoning, not only recall.

Assessment approach. Students are expected to participate in scenario analysis, complete assigned knowledge checks, capture required lab evidence, and contribute to remediation planning discussions. The instructor may adapt the assessment depending on whether the class is delivered as an executive workshop, technical implementation class, or self-paced online module.

Component	Purpose	Typical evidence
Knowledge checks	Validate understanding of concepts, terminology, permissions, and Defender capabilities.	Quiz results or instructor-reviewed answers.
Lab tasks	Demonstrate ability to navigate portals, inspect posture, and perform approved configuration steps.	Screenshots, notes, resource names, and brief explanations.
Scenario discussion	Connect business risk to technical controls and remediation priorities.	Group notes, threat mapping, and prioritization decisions.
Operational review	Summarize environment findings and recommended next actions.	Short remediation plan or action register.
Professional conduct	Operate safely within scope and respect data handling rules.	No unauthorized changes or sensitive data exposure.

Evidence capture standard

- Use clear screenshots that show the relevant portal area, recommendation, configuration, or validation result.
- Do not expose secrets, access keys, tokens, passwords, private endpoints, customer personal data, or confidential identifiers.
- Add a short caption explaining what the screenshot proves.
- Record the resource group, resource type, and action performed where applicable.
- Submit evidence using the LMS, Teams, Loop, GitHub issue, or mechanism specified by the instructor.

Suggested pass criteria

A practical pass should require attendance, active participation, completion of assigned knowledge checks, and acceptable lab evidence or scenario output. Formal grading rules should be provided separately in the student assessment rubric.

SECTION 10

Recommended Student Study Workflow

Use this workflow to prepare, participate, and convert the workshop into practical follow-through.

Phase	Recommended activity	Output
5-7 days before class	Complete account checks, MFA, portal	Readiness confirmed.

	access, Microsoft Learn sign-in, and pre-work reading.	
1-2 days before class	Verify assigned Azure subscription, resource group, roles, and network access. Confirm no production data will be used.	Lab access confirmed.
During class	Follow the instructor, take notes, ask scope questions, capture evidence, and complete activities in sequence.	Lab evidence and course notes.
Same day after class	Clean up lab resources, disable unneeded paid plans, and record open questions.	Cleanup completed.
Within 7 days	Complete remaining Microsoft Learn modules and rewrite remediation findings into an action plan.	Post-course action register.
Within 30 days	Review production posture with proper approval and convert recommendations into backlog items.	Prioritized security roadmap.

Suggested self-paced completion path

11. Refresh fundamentals using SC-900-aligned security, compliance, identity, Entra, Microsoft security solutions, and Purview modules.
12. Complete Defender for Cloud posture and threat mitigation modules relevant to your role.
13. Complete AI security modules if your organization uses Azure AI, Microsoft Foundry, Copilot, or generative AI workloads.
14. Review SC-100 architecture content if you are responsible for strategy, Zero Trust design, governance, or security architecture.

SECTION 11

Student Action Register Template

Use this lightweight register to convert course findings into practical remediation work.

Finding / opportunity	Risk or business impact	Recommended action	Owner	Priority	Target date

How to prioritize

Start with internet-facing exposure, identity over-permissioning, missing protection plans for critical workloads, vulnerable assets, regulatory gaps, and AI workloads that process sensitive or business-critical data.

SECTION 12

Glossary of Key Terms

These terms appear frequently in the course, labs, and Microsoft Learn resources.

Term	Meaning in this course
CSPM	Cloud Security Posture Management. Visibility, recommendations, risk prioritization, secure score, compliance, and attack path analysis.
CWPP	Cloud Workload Protection Platform. Runtime and workload-specific protection for servers, storage, databases, containers, AI services, and other assets.
Secure Score	A posture indicator that helps prioritize security recommendations and improvement actions.
Attack path	A linked sequence of exposures or weaknesses that could allow an attacker to reach a high-value asset.
Security Exposure Management	A risk-focused approach to understanding exploitable paths, exposure, and mitigation priorities.
Microsoft Defender portal	Unified security operations portal for incidents, alerts, investigation, correlation, and response across Microsoft security products.
Microsoft Defender for Cloud	Microsoft platform for cloud security posture management and workload protection across Azure, hybrid, and multicloud environments.
Microsoft Entra ID	Cloud identity and access management platform used for

	users, workloads, authentication, authorization, roles, and Conditional Access.
Microsoft Foundry	Microsoft platform context for building and operating AI solutions; course content references AI workload protection and identity controls.
RBAC	Role-Based Access Control. Determines which users or identities can perform actions at subscription, resource group, or resource scope.
MFA	Multifactor Authentication. A control that requires more than a password to sign in.
DLP	Data Loss Prevention. Policies and controls used to reduce data leakage risk.
EASM	External Attack Surface Management. Discovery and assessment of unknown or internet-facing assets.
MCRA	Microsoft Cybersecurity Reference Architecture. Architecture guidance referenced in SC-100-aligned learning.
CAF / WAF	Cloud Adoption Framework and Well-Architected Framework. Microsoft frameworks used to guide governance, architecture, and operational excellence.

APPENDIX A

Student Pre-Course Checklist

Complete this checklist before the course starts.

Done	Checklist item
☐	I can sign in to my Microsoft Learn account.
☐	I can access the Azure portal from my class device and network.
☐	I can see the correct Microsoft Entra tenant/directory.
☐	MFA is configured and tested.
☐	I know which Azure subscription or resource group is assigned to me.
☐	I know my assigned permission level: Reader, Contributor, Security Admin, or other.
☐	I can access Microsoft Defender for Cloud where required.
☐	I can access Microsoft Defender portal where required.
☐	My laptop/desktop, browser, internet connection, PDF reader, and screenshot tool are ready.
☐	I understand that paid Defender plans may create cost and must only be enabled inside approved scope.
☐	I will not use production, personal, customer, HR, legal, financial, or confidential data in labs.
☐	I understand the cleanup expectations after the labs.

APPENDIX B

Portal and Access Checklist

Use this during class if access issues occur.

Portal / service	What to confirm
Microsoft Learn	Sign-in works and the course collection is accessible.
Azure portal	Correct tenant selected; assigned subscription/resource group visible.
Microsoft Defender for Cloud	Environment settings, recommendations, Secure Score, inventory, workbooks, and compliance views load.
Microsoft Defender portal	Incidents, alerts, and XDR views are accessible if included in the course track.
Microsoft Entra admin center	Visible only if identity labs or administrative review are included.
GitHub or course site	Courseware, lab files, or self-paced content can be accessed.

Teams / Loop / LMS	Course communication, submissions, and checklists are accessible.
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APPENDIX C

Lab Naming and Cleanup Checklist

Use consistent naming and clean up after the workshop to reduce cost and residual access risk.

Recommended naming pattern

Resource type	Example naming pattern
Resource group	rg-cese323-lab-<studentname>
Storage account	stcese323<random>
App Service	app-cese323-<studentname>
AKS cluster	aks-cese323-<studentname>
AI resource	ai-cese323-<studentname>

Cleanup checklist

Item	Cleanup action
Defender paid plans	Disable after lab unless the customer or lab owner explicitly wants them retained.
AKS clusters	Delete after the lab if no longer required.
Storage accounts	Delete after evidence is captured and approved.
App Services	Stop or delete after lab completion.
AI resources/model deployments	Delete, disable, or scale down after lab as instructed.
Role assignments	Remove temporary student access.
Budgets and alerts	Review spend after the course.
Evidence files	Store only sanitized, approved evidence in the required submission location.

APPENDIX D

Student Notes

Use this space for environment-specific notes, instructor guidance, and follow-up actions.

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- 2.
- 3.
- 4.
- 5.

6.

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APPENDIX E

Course Source References

Reference sources used to assemble this student guide.

Primary courseware sources

- Cloud Security Envisioning & Strategy Workshop Syllabus AZ.docx - workshop objectives, agenda, workload selection, practical remediation, and operational review structure.
- Pre-Course Lab Requirements.txt - student readiness, account access, permissions, lab safety, workload prerequisites, live environment controls, and cleanup expectations.
- Microsoft-Learn-Collection.txt - Microsoft Learn collection URL and collection context.

Microsoft Learn and documentation links captured in the course source pack

- [Cloud Security Envisioning & Strategy Workshop Syllabus collection](#)
- [Defender XDR prerequisites](#)
- [Connect Azure subscriptions to Microsoft Defender for Cloud](#)
- [Microsoft Defender for Cloud permissions](#)
- [Defender for Storage support matrix](#)
- [Enable Defender for Containers](#)
- [AI threat protection onboarding](#)
- [Implement Defender for Cloud AI services training module](#)

End of Student Course Guide